

TASK 3 : THE KNOWLEDGE ANALYST

(RAG CONCEPTS)

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Objective

The objective of this project is to simulate a Retrieval-Augmented Generation (RAG) workflow by analyzing a long document using AI. The system is designed to answer questions based only on the uploaded document, provide references, and extract important information such as risks, dates, and stakeholders.

Tools Used

- Claude AI
- Pdf Document
- Microsoft Word

Document Used

Name of document : environmental science and pollution

Description:

A document was uploaded to Claude AI and used as the knowledge source for question answering and information extraction.

Prompt Used

You are a document analyst.

Answer every question using ONLY information found in the uploaded document.

For every answer:

Provide the answer.

Mention the page number.

Mention the section heading if available.

If the information is not present in the document, say:

"Information not found in the document."

Question 1 : what are the sources of air pollution ?

Question 2 : what is solid waste management and what are its classifications ?

Sources of Air Pollution

Page: 3 | Section: Air Pollution

The document divides the sources of air pollution into two categories:

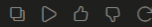
Natural Sources

1. Volcanic eruptions (SO_2 , H_2S)
2. Decay of vegetation
3. Marsh gases
4. Pollen grains
5. Forest fires

Man-made Sources

1. Deforestation
2. Fossil fuel combustion
3. Industrialisation
4. Use of pesticides & insecticides
5. Explosives used in war

Definition (Page 3): Air Pollution is defined as "the presence of chemicals in the atmosphere in quantities and durations that are harmful to human health and the environment."



Classification of Solid Waste

The document lists 6 types of solid waste:

#	Type	Description
1	Garbage	Biodegradable organic waste from food
2	Rubbish	Dry, non-putrescible material (paper, glass, metal)
3	Pathological Waste	Hospital and biological waste
4	Industrial Waste	From various Industrial processes
5	Agricultural Waste	Crop residues, animal manure
6	Ashes	Residue from combustion of solid fuels

Process Flow of Solid Waste Management

Process Flow of Solid Waste Management

(Also shown on Page 16 as a diagram)

The management process follows these steps in order:

Collection → Transportation → Recovery → Processing → Disposal

Solid Waste Management & Its Classifications

Page: 16 | Section: Solid Waste Management

Definition

Solid Waste Management is defined as:

"The discipline associated with the control of generation, storage, collection, transport, processing, and disposal of solid waste materials."

Classification of Solid Waste

Question 3: what is the difference bw floods and earthquakes?

Question 4: what were the three major nuclear accident?

what is the difference bw floods and earthquakes?

Floods vs Earthquakes

Page: 23 | Section: Managing Natural Disasters: Floods & Earthquakes

Floods

Aspect	Details
Definition	A huge amount of water submerging normally dry land
Types	Flash floods, Coastal floods, and Urban floods
Mitigation	Learn warning signs, collect emergency supplies, have evacuation routes planned, and install check valves in sewer traps

Earthquakes

Aspect	Details
Definition	Result of a sudden release of energy in the Earth's crust
Key Terms	Hypocenter is the origin point; Epicenter is the surface point directly above
Mitigation	Build earthquake-resistant structures, identify safe places indoors/outdoors, and have a family communication plan

Key Differences at a Glance

Feature	Floods	Earthquakes
Cause	Excess water submerging land	Sudden release of energy in Earth's crust

es

Cause	Excess water submerging land	Sudden release of energy in Earth's crust
Types mentioned	Flash, Coastal, Urban	Not categorized into types
Unique concept	Evacuation routes & check valves	Hypocenter & Epicenter
Mitigation focus	Warning signs & emergency supplies	Earthquake-resistant structures & communication plan

All information sourced exclusively from Page 23 of the document.

what were the three major nuclear accident?

Three Major Nuclear Accidents

Page: 25 | Section: Nuclear Accidents

Definition of a Nuclear Accident

"An event that has led to significant consequences to people, the environment, or the facility. A major accident involves damage to the reactor core and a significant release of radioactivity."

Three Major Incidents

#

Incident

Location

Year

Three Major Incidents

#	Incident	Location	Year
1	Three Mile Island	USA	1979
2	Chernobyl	USSR	1986
3	Fukushima Daiichi	Japan	2011

All information sourced exclusively from Page 25 of the document. The document does not provide further details about each individual accident beyond their name, location, and year.

Question 5 : what is the purpose of document?

what is the purpose of document

The document does not explicitly state its purpose anywhere in the content.

However, based on the information present in the document, it can be reasonably inferred that it is:

An Educational Study Material for EVS (Environmental Studies) — Unit 2, created by Bunktobrain (bunktobrain.com), intended to help engineering or college students prepare for their exams by covering topics like Pollution, Waste Management, Disaster Management, and more.

Evidence from the document itself (Pages 1 & 26):

Management, Disaster Management, and more.

Evidence from the document itself (Pages 1 & 26):

- The cover page (Page 1) shows the Bunktobrain branding and mentions:
 - "For Free lectures, Notes, Paper leak, PQS, Most Imp Questions!"
- The last page (Page 26) says:
 - "All the Best for your Exams!"
 - "For Notes, Paper leak, PQS and Most Imp Questions!"

⚠ Note: The document does not contain a formally written "purpose" or "objective" section. The above is based only on what is visible within the document itself.

Results:

The AI successfully extracted information from the uploaded document and provided answers with references to the relevant sections. This demonstrates the basic principles of Retrieval-Augmented Generation (RAG), where responses are grounded in source documents rather than generated solely from the model's general knowledge.

Conclusion:

This project successfully simulated a RAG-based document intelligence workflow. The AI was able to analyze a lengthy document, answer questions accurately, provide references, and generate a structured summary dashboard containing risks, dates, stakeholders, and key insights.